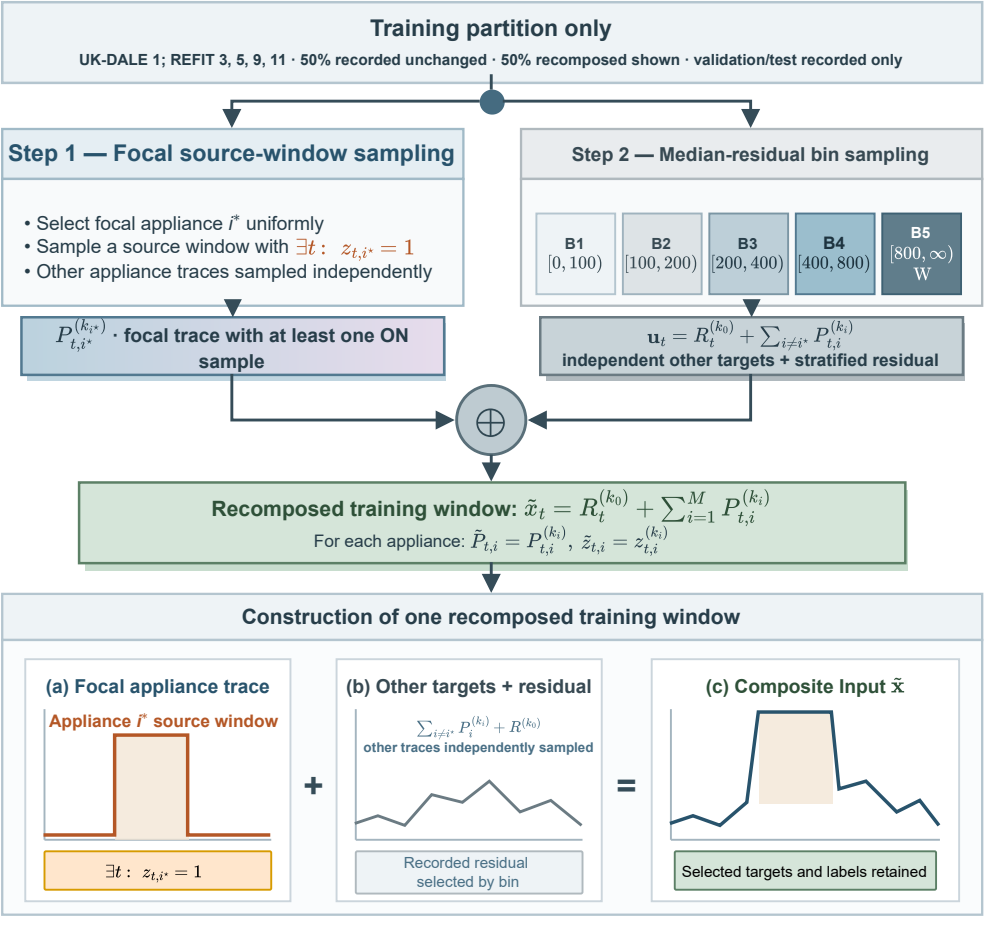

Training partition only
UK-DALE 1; REFIT 3, 5, 9, 11 · 50% recorded unchanged · 50% recomposed shown · validation/test recorded only
Step 1 — Focal source-window sampling
• Select focal appliance i^* uniformly
• Sample a source window with $\exists t: \; z_{t,i^*} = 1$
• Other appliance traces sampled independently
Step 2 — Median-residual bin sampling
B1 [0, 100)
B2 [100, 200)
B3 [200, 400)
B4 [400, 800)
B5 [800, ∞) W
$P_{t,i^*}^{(k_{i^*})}$ · focal trace with at least one ON sample
$\mathbf{u}_t = R_t^{(k_0)} + \sum_{i \neq i^*} P_{t,i}^{(k_i)}$
independent other targets + stratified residual
Recomposed training window: $\tilde{x}_t = R_t^{(k_0)} + \sum_{i=1}^{M} P_{t,i}^{(k_i)}$
For each appliance: $\tilde{P}_{t,i} = P_{t,i}^{(k_i)}$, $\tilde{z}_{t,i} = z_{t,i}^{(k_i)}$
Construction of one recomposed training window
(a) Focal appliance trace
Appliance i^* source window
$\exists t: \; z_{t,i^*} = 1$
+
(b) Other targets + residual
$\sum_{i \neq i^*} P_i^{(k_i)} + R^{(k_0)}$
other traces independently sampled
Recorded residual selected by bin
=
(c) Composite Input $\tilde{\mathbf{x}}$
Selected targets and labels retained